

N-Gram Language model

GOAL

- compute the probability of a sentence or sequence of words

$$P(w) = P(w_1, w_2, w_3, \dots, w_n)$$

RELATED TASK

probability of an upcoming word

$$P(w_4 | w_1, w_2, w_3)$$

A model that computes either of these is called a language model

Example:

we want to compute probability of the following sequence of words

"about fifteen minutes from"

i.e. $P(\text{about, fifteen, minutes, from})$

- compute using chain rule of probability

$$P(B|A) = \frac{P(A, B)}{P(A)}$$

$$\therefore P(A, B) = P(B|A) P(A)$$

similarly,

$$P(A, B, C) = P(A) P(B|A) P(C|A, B)$$

$$P(A, B, C, D) = P(A) P(B|A) P(C|A, B) P(D|A, B, C)$$

In general,

$$P(x_1, x_2, \dots, x_n)$$

$$= P(x_1) P(x_2|x_1) P(x_3|x_1, x_2) \dots P(x_n|x_1, \dots, x_{n-1})$$

$$\therefore P(w_1, w_2, \dots, w_n) = \prod_i P(w_i | w_1, w_2, \dots, w_{i-1})$$

$$\therefore P(\text{"about fifteen minutes from"})$$

$$= P(\text{about}) \times P(\text{fifteen} | \text{about}) \times$$

$$P(\text{minutes} | \text{about fifteen}) \times$$

$$P(\text{from} | \text{about fifteen minutes})$$

$$\text{Calculating, } P(\text{office} | \text{about fifteen minutes from})$$

$$= \frac{\text{count}(\text{about fifteen minutes from office})}{\text{count}(\text{about fifteen minutes from})}$$

Problem: might not have enough data for estimating these values

So simplification required

Markov Assumption

(3)

① Use only the previous word

eg: $P(\text{office} | \text{about fifteen minutes from}) \approx P(\text{office} | \text{from})$

② or use couple of ~~prev~~ previous words

eg: $P(\text{office} | \text{about fifteen minutes from}) \approx P(\text{office} | \text{minutes from})$

∴ Instead of chain rule use kth order markov model

kth order markov model

Chain Rule:

$$P(w_1 w_2 \dots w_n) = \prod_i P(w_i | w_1 w_2 \dots w_{i-1})$$

using markov Assumption: use only k previous words

$$\therefore P(w_1 w_2 \dots w_n) \approx \prod_i P(w_i | w_{i-k} \dots w_{i-1})$$

Thus we approximate each component in the product

$$\therefore P(w_i | w_1 w_2 \dots w_{i-1}) \approx P(w_i | w_{i-k} \dots w_{i-1})$$

N-gram language models / N-gram models

An N-gram model uses only N-1 words of prior context

Eg: For $P(\text{office} | \text{about fifteen minutes from})$

Unigram : $P(\text{office})$

Bigram : $P(\text{office} | \text{from})$

Trigram : $P(\text{office} | \text{minutes from})$

This can be ~~extended~~ extended to

4-grams, 5-grams

- (4)
- Relation between Markov Model and Language Model

N -gram model = $N-1$ order Markov model

Limitations of Language Model

- cannot say that n -gram model of language is sufficient because language has long distance dependencies.

eg: "The computer which I had just put into the machine room on the fifth floor crashed"

Calculation of N-gram Probabilities

- It is based on

Maximum Likelihood Estimate

$$P(w_i | w_{i-1}) = \frac{\text{count}(w_{i-1}, w_i)}{\text{count}(w_{i-1})}$$

$$P(w_i | w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

Limitations

- underflow due to multiplication of probabilities
solution: use log. i.e. $\log(p_1 \times p_2 \times p_3) = \log(p_1) + \log(p_2) + \log(p_3)$
- zero probability problem / Handling zeros
solution: use Smoothing - we'll use Laplace smoothing

Example:

a) Calculate Bigram Probabilities for the given corpus

Corpus Data:

<S> I am here <S>

<S> who am I <S>

<S> I would like to know <S>

Solution:

- To compute Bi-gram probabilities,

$$\text{use } P(w_i | w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

- Estimating probabilities for bigrams

$$P(I | <S>) = 2/3$$

$$P(<S> | here) = 1/1$$

$$P(would | I) = 1/3$$

$$P(here | am) = 1/2$$

$$P(know | like) = 0/1 = 0$$

- Estimating bigram probabilities of the corpus data

For it, list down all unique words from corpus including start of sentence and end of sentence

Bigram probabilities for corpus data

(6)

$w_{i-1} \backslash w_i$	<s>	I	am	here	who	would	like	to	know	</s>
<s>	0	$\frac{2}{3} = 0.66$	0	0	$\frac{1}{3} = 0.33$	0	0	0	0	0
I	0	0	$\frac{1}{3} = 0.33$	0	0	$\frac{1}{3} = 0.33$	0	0	0	$\frac{1}{3} = 0.33$
am	0	$\frac{1}{2} = 0.5$	0	$\frac{1}{2} = 0.5$	0	0	0	0	0	0
here	0	0	0	0	0	0	0	0	0	1
who	0	0	$\frac{1}{1} = 1$	0	0	0	0	0	0	0
would	0	0	0	0	0	0	$\frac{1}{1} = 1$	0	0	0
like	0	0	0	0	0	0	0	$\frac{1}{1} = 1$	0	0
to	0	0	0	0	0	0	0	0	$\frac{1}{1} = 1$	0
know	0	0	0	0	0	0	0	0	0	$\frac{1}{1} = 1$
</s>	$\frac{2}{3} = 0.66$	0	0	0	0	0	0	0	0	0

Q2) For corpus given in Q1, using bigram probabilities, compute probability of sentence
 <s> who would like to ~~know~~ know </s>

Solution: $P(\text{<s> who would like to know </s>})$
 $= P(\text{who} | \text{<s>}) \times P(\text{would} | \text{who}) \times P(\text{like} | \text{would}) \times$
 $P(\text{to} | \text{like}) \times P(\text{know} | \text{to}) \times P(\text{</s>} | \text{know})$
 $= \frac{1}{3} \times 0 \times 1 \times 1 \times 1 \times 1$

Handling zero probability value using Laplace Smoothing

Laplace Smoothing	
1) calculate frequency of unique words	
word	<S> I
freq	

Laplace Smoothing

- 1) count the total number of unique words in corpus
unique words =

<S>, I, am, here, who, would, like, to, know, <I>

Total count = 10

- 2) Exclude <S> (start of sentence) from total count

$$\text{count of unique words without } \langle S \rangle = 9$$

- 3) For every probability value, update

$$\text{numerator} = \text{numerator} + 1$$

$$\text{denominator} = \text{denominator} + \text{count of unique words without } \langle S \rangle$$

Applying Laplace Smoothing

$P(<S> \text{ who would like to know } <Is>)$

$$= 1/3 \times \underline{0} \times 1 \times 1 \times 1 \times 1$$

After Laplace Smoothing

$P(<S> \text{ who would like to know } <Is>)$

$$= (1+1)/(3+9) \times (0+1)/(1+9) \times (1+1)/(1+9) \times$$

$$(1+1)/(1+9) \times (1+1)/(1+9) \times (1+1)/(1+9)$$

$$= 2/12 \times 1/10 \times 2/10 \times 2/10 \times 2/10 \times 2/10$$

$$= 0.16 \times 0.1 \times 0.2 \times 0.2 \times 0.2 \times 0.2$$

$$= 0.000256$$

$$P(<S> \text{ who would like to know } <Is>) = 0.000256$$

Q3) For corpus given in Q1, using bigram probabilities, find which of the given sentence is better (i.e. find the sentence having higher probability)

1) $<S> \text{ who would like to know } <Is>$

2) $<S> \text{ who would I } <Is>$

Practice Question ↗